How email becomes answers your AI assistants can cite

Company email is read from Microsoft 365, screened and classified by AI inside the Finishes Azure tenant, stored in one PostgreSQL database, and served to Claude, ChatGPT and Copilot Studio through a single authorized endpoint. Each person signs in as themselves and sees only the mail they are allowed to read.



Two timers, one queue, one atomic publish

Everything runs inside the ingestion Function App on two 5-minute timers. Capture only stores mail; processing turns stored mail into searchable, classified evidence. Each message is processed in one database transaction, so a crash leaves it queued for the next run, never half-published.

LANE 1 — Capture

Function `capture`

01 — Timer tick
Every 5 minutes the ingestion app wakes up.
FREE

02 — Is a capture due?
Reads the schedule (on/off, interval, look-back window, active hours, weekdays) and the mode: **test**, **production** or **paused**.
FREE

03 — Which mailboxes
Enrolled, approved and not excluded. Exclusions always win. An empty scope stops the run.
FREE

04 — Read new mail
Microsoft Graph, read-only, immutable IDs. Skips Deleted, Junk, Drafts, Outbox. Up to 93 days on request.
GRAPH

05 — Store
Message, participants and attachment details; each mailbox copy kept once. Queued as **fetched**.
FREE

LANE 2 — Process

Function `process`

01 — Claim
4 workers take queued messages, each in its own transaction, within a time budget.
FREE

02 — Attachments
Text from PDF, Word, Excel and text files in an isolated process with size and time limits; original kept in Blob.
FREE

03 — Bulk pre-filter
No-reply senders and unsubscribe mail are skipped before any paid step.
FREE

04 — Prompt Shields
Looks for prompt-injection in what the classifier will read.
PAID · SWITCH

05 — Masking
IDs, bank, routing and card numbers, passwords, keys and tokens become [Category] in stored text.
PAID · SWITCH

06 — Classify
One model call: 21 labels, summary, action, due date, project risk, properties, entities and knowledge.
PAID

07 — Decide
Restricted, not indexed, review or indexed, using the label thresholds.
FREE

08 — Person check
For each person profile (e.g. Chuck Sr. vs Jr.): is this really their action item?
PAID

09 — Knowledge
Names, people, facts and terms added to the knowledge base, checked against the email.
FREE

10 — Embed & publish
~350-token chunks, 1,536-number vectors, then published in one atomic step.
PAID

OUTCOMES

What happens to each message

OUTCOME	SEARCHABLE	WHEN
INDEXED	Yes	Normal business mail
REVIEW	Yes, flagged	Phishing not ruled out; masked IDs or credentials; uncertain
NOT INDEXED	No	Spam, newsletters, bulk and automated senders
RESTRICTED	Never	Possible health information (and IDs or credentials when masking is off)
HELD	Not yet	A screening or masking service failed; retried, never passed

SWITCHES

Who turns what on and off

SWITCH	WHERE	EFFECT
CAPTURE_ENABLED, **PROCESSING_ENABLED**	Azure app settings	Hard kill switches
Mode: test / production / paused	`ci mode set`	Paused stops capture and publishing
Schedule: interval, window, hours	Console · Schedule & runs	When capture runs; run now
Prompt Shields, masking	Console · Schedule & runs	Off saves money while testing; not for production
Chat model	Console · Costs	Tested on a synthetic email before switching

COST

How spend is tracked

- Every paid call is metered per run: OpenAI tokens as reported, Language and Content Safety per 1,000 characters.
- Estimated cost uses editable prices (unverified list prices until checked against the bill).
- Billed cost comes from Azure Cost Management, 8–24 hours behind, side by side on the Costs page.
- Free steps run first, so skipped mail costs nothing.

One door for every AI assistant, with the person's own permissions

Assistants never get a database connection, key or file. They call the search app's MCP endpoint with the person's own Microsoft sign-in; the endpoint checks the token, applies that person's grants in the database before anything is ranked or counted, and records the call.

AGENT CLIENTS

How each assistant connects

CLIENT	HOW IT CONNECTS	STATUS
ChatGPT	Custom MCP plugin, user-defined OAuth client, Resource = api://<API id>, PKCE on, OIDC off	VERIFIED
Claude	Admin settings → Connectors → custom connector; OAuth client ID and secret	PER GUIDE
Copilot Studio	Tools → Model Context Protocol, manual OAuth, end-user credentials	PER GUIDE

Entra has no dynamic client registration, so each client has its own pre-registered app. "Per guide" means set up as documented but not yet recorded as verified.

READER ROLES

What a person can see

ROLE	SEES
Communications.Reader.All	Every enrolled mailbox
Communications.Reader.Own	Their own mailbox, plus explicit grants
Communications.Admin	The admin console and the /read email and file links

Managed with `scripts/azure/people.sh`. Every grant to read all mail is an access-review item.

MCP TOOLS · /MCP

11 tools, one authorization path

TOOL	USE IT FOR
`search_email`	Find emails by meaning and keywords; filters for mailbox, sender, label, property, entity, dates
`list_email`	Newest emails matching filters
`list_actionable`	Action items, approvals and project risks; per person with the person check
`get_email`	One email: people, labels, AI summary, attachments
`get_attachment`	An attachment's extracted text, a page at a time
`get_thread`	The whole conversation, oldest first
`get_file_links`	Email and file links for up to 25 emails, to put on a board
`email_stats`	Counts by mailbox, sender, domain, label, property, entity or period
`risk_overview`	Project-risk trends, top properties and companies, newest risks
`find_knowledge`	Look up a property, company, person or term
`get_profile`	Everything about one entry: facts, related entries, activity, emails

IDENTITIES

Least privilege by design

IDENTITY	CAN	CANNOT
Ingestion	Read mail (Exchange RBAC), write the database and Blob, call Azure AI	Send or change mail
Search	Read published views only	Read mail, write anything
Admin console	Operator settings, published views, read Blob, read costs	Read mail
GitHub deploy	Deploy the Function Apps (OIDC, no stored secret)	Read data

MONDAY.COM

Putting email and files on a board

1. **Ask the assistant**, e.g. "put the files from the Acme invoice on the Acme intake item".
2. **get_file_links** returns the email link and each kept file's link.
3. The assistant's **own monday connector** adds them as links, under your monday sign-in.

No direct monday integration and no file copies. Links open only for console admins. A Files column taking links is not officially documented by monday; a Link column is the fallback.

ONE QUESTION, END TO END

What happens when someone asks an assistant

01 Ask — A person asks Claude, ChatGPT or Copilot, e.g. “which properties had lien threats this month?” — PERSON

02 Call a tool — The assistant picks a tool (search_email, risk_overview…) and sends the person's own Entra token. — MCP

03 Check the token — Signature, tenant, audience, expiry and the Communications.Search scope; otherwise 401. — SEARCH APP

04 Resolve access — Reader.All, Reader.Own or explicit grants, from the token's roles; never from tool arguments. — SEARCH APP

05 Search — Grants applied in SQL first; then vector top 50 + full-text top 50, fused (RRF). — POSTGRESQL

06 Collapse copies — One message in several mailboxes is shown once, after authorization. — SEARCH APP

07 Audit — Who, which tool, which result IDs; never the question text. — POSTGRESQL

08 Return evidence — Excerpts, AI summary, labels, citations and a view_url per email; text marked untrusted. — SEARCH APP

09 Answer — The assistant answers and cites sender, date and subject for each claim. — ASSISTANT

10 Open the source — Admins open view_url to read the email or download its files after Microsoft sign-in. — ADMIN CONSOLE

What is stored, how it learns, how it ships

One PostgreSQL database holds the queue, the evidence, the search index and the knowledge base. Search reads only “published” views, so mail that is held, restricted, excluded or withdrawn is unreachable. Code ships through GitHub; database changes are applied first, from an approved machine.

DATABASE

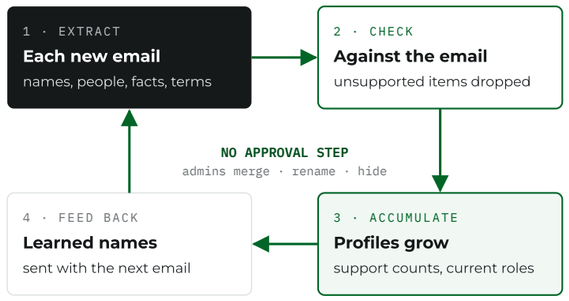
What lives where

AREA	HOLDS
Scope & control	Mailboxes, approvals, exclusions, mode, schedule, switches
Evidence	Messages, versions, participants, attachments, masked text
Search index	Chunks with 1,536-d vectors (HNSW) and full-text
Classification	Labels with probabilities, summary, action, risk, names
Action items	Person profiles and per-person checks
Knowledge	Entities, mentions and facts, per published message
Feedback	Reviewer answers: precision and recall per label
Operations	Runs, AI usage, prices, audit log

Roles: `ci_ingest` writes, `ci_search` reads published views, `ci_operator` runs settings. Migrations 0001–0016.

KNOWLEDGE BASE

Learning from email over time



- Properties, companies, people (by address), terms and facts: works for, role on a property, owns, affiliate of.
- Phishing-flagged, restricted and unindexed mail add nothing; no payment details are stored.
- Every fact shows how many emails and senders support it; counted only from mail the reader may see.

GUARDRAILS

Rules the system enforces

- Email text is untrusted: instructions inside it are ignored.
- No secrets or email content in Git, logs or test data; logs hold IDs, counts and codes.
- Permissions are applied in the database before ranking; caller filters only narrow.
- Never tenant-wide Mail.Read: Exchange RBAC is the mailbox boundary.
- AI output is a proposal, never an approval; thresholds change only by a person.
- Every read and every admin change is audited.

SHIP & OPERATE

From a change to running code

1. **Pull request**: lint, type check, tests on PostgreSQL 18 + pgvector 0.8.2, package build.
2. **Database first**: `make azure-env`, then `make azure-migrate` from the Mac.
3. **Merge**: Deploy (development) builds and deploys only the apps that changed, via GitHub OIDC.
4. **Operate**: console for schedule, mailboxes, costs, review; `ci` for scope, mode, backfill.

ACCEPTANCE · CLAUDE.MD §11

How “works 95% of the time” will be measured: five separate numbers

CAPTURE
Completeness
Mail in each mailbox versus mail captured, per mailbox, with counts.
NOT MEASURED YET

PROCESSING
Success
Published, held by reason, failed and dead; oldest waiting message.
NOT MEASURED YET

CLASSIFICATION
Label quality
Precision and recall per label on hand-checked mail, with sample sizes.
NOT MEASURED YET

RETRIEVAL
Answer quality
Recall@5/@10 and MRR on 30–50 questions with known answers, including ones with none.
NOT MEASURED YET

AUTHORIZATION
Negatives
Wrong audience, missing role, revoked grant, excluded mailbox, paused mode, forged filter: zero results.
PARTLY TESTED (SYNTHETIC)

BUILT AND TESTED ON SYNTHETIC MAIL

CAPTURE & PROCESSING · MASKING & SCREENING · CLASSIFICATION & ACTIONS · PERSON CHECK · HYBRID SEARCH · 11 MCP TOOLS · ADMIN CONSOLE · COST TRACKING · KNOWLEDGE BASE · FILE LINKS

NOT BUILT OR NOT YET MEASURED

PER-FOLDER DELTA & WEBHOOKS · DELETES & MOVES AFTER CAPTURE · REST ENDPOINTS · BICEP INFRASTRUCTURE · MALWARE SCANNING & OCR · LABEL CALIBRATION · KNOWLEDGE FACT QUALITY · PRODUCTION ENVIRONMENT · TEAMS, FIREFLIES, DIALPAD